

EXPERIMENT — 15

AIM: Project Demo and Report Submission with the team

REQUIREMENTS: Windows 10/11, Chrome Browser, Docker, VS Code.

Features Demonstrated in Final Demo:

1. **User Authentication & Secure Login:** Users log in with credentials and are redirected to a role-specific dashboard.
2. **Role-Based Access Control (RBAC):** Different users (admin, warehouse staff, driver, manager) see only relevant modules and data.
3. **Centralized Dashboard:** Unified dashboard showing orders, inventory, logistics, and finance overview in one place.
4. **Order Management System:** Create, update, and track manufacturer & customer orders end-to-end.
5. **Inventory Management with Real-Time Updates:** Tracks stock across warehouses with bin-level accuracy and alerts (e.g., low stock).
6. **Warehouse Management:** Manage warehouse data, stock placement, and movement of goods efficiently.
7. **Delivery & Logistics Management:** Assign drivers and vehicles, track delivery status (Assigned → Dispatched → Delivered).
8. **Activity Logs & Monitoring:** Tracks user actions (login, updates, deletions) for accountability and auditing.
9. **Responsive UI & Multi-Module Navigation:** Clean UI with modules like supply chain, logistics, warehouse, etc. (shown in UI pages).
10. **End-to-End Workflow Execution:** Complete flow: Login → Create Order → Inventory Check → Assign Delivery → Delivery Completion

Key Technical Achievements:

1. **Full-Stack Architecture Implementation**
Built using React (frontend) + Node.js backend + database integration.
2. **Modular System Design**
Separate modules for orders, inventory, logistics, finance → easy scalability and maintenance.
3. **Transactional Database Operations**
Used atomic transactions (e.g., delivery creation updates multiple tables safely).
4. **Role-Based Security System**
Implemented RBAC with controlled access to modules and sensitive data.

5. **Real-Time System Behaviour**

Inventory, delivery, and order status updates dynamically across system modules.

6. **Comprehensive Testing Strategy**

Includes unit, integration, system testing and UAT with all test cases passing.

7. **Scalable & Performance-Oriented Design**

Supports multiple users with optimized queries and modular backend.

8. **Secure Data Handling**

Includes authentication, encryption concepts, logging, and session management.

Version History:

Version	Date	Implementation Status
V1.0	January 2026	Core Setup Completed
V1.1	February 2026	Core Modules Built
V1.2	March 2026	Advanced Features Integrated
V2.0	April 2026	Testing & Deployment Ready

RESULT:

Project successfully demonstrated and report submitted.